A QUANTITATIVE STUDY OF AI-VOICE BASED ASSISTANT USERS THAT ANTHROPOMORPHIZE TECHNOLOGY AS SOCIAL ACTORS

by

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Abstract

The problem was that there were limited studies that acknowledged how voice-based assistant users perceive the intelligence of voice-based assistants and the different variation levels of trust towards it. The purpose of this quantitative research was to investigate how demographic variables such as generation, ethnicity, gender and education level influence perceptions of trust and perceive intelligence towards AI-voice based assistants. The rationale for this research was that voice-based assistants have garnered 46% of the human-technology interaction. It is pivotal that voice-assistant applications listen to the need of its users' as it pertains to the user experience. Voice-based assistants users' perception acts as a microscope that speaks volume to voice assistants' ability and how it gets culturally utilized. Recent studies have shown that voice-based assistant user' perception of intelligences dictates users' attitude towards the technology; also, their perception of anthropomorphism shifts. The central research question for this study was: Do human beings trust voice-based assistants more than themselves? The quantitative design used a survey that consisted of various closed-ended questions such as the 7-point Likert scale. Amazon M-Turk was utilized to expose a simple random sampling technique, and a total of 1171 qualified participants completed the survey. Data were analyzed through a series of descriptive statistics, ANOVA, an independent t-test and linear regressions to analyze the other mentioned research questions.

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Dedication

I dedicate this dissertation to God, my grandmother, my younger sister, my parents, the committee, Dr. Mott, participants and all those who have supported me during this incredible journey. Your love and genuine support have been fulfilling in securing confidence to achieve academic excellence at the highest level. To my grandmother, I dedicate this to you. You have been unequivocal a strong supporter of me in my academic ability and willingness to succeed despite my own insecurities. You have witnessed my trials and tribulations; yet, continued to show up for me even when I did not possess the necessary confidence to withstand some of life's over bearings. I am grateful for you being the example of what it means to proclaim consistency and resiliency as God as the leader of our lives. To my younger sister Natasha Oliveira, thank you so much for supporting me. Your kind words and family have been a true form of authenticity in giving me hope. I appreciate you for showing me what love is and the family bond that it creates to show what success can look like. To the committee, I would like to express my deep appreciation and gratitude for your high level of expertise and constructive feedback to help me grow in stimulating my mind. I am very honored to have had the opportunity to gain insight and become an active intentional learner. To Dr. Mott, thank you for guiding me and being a pivotal leader for me in not only demonstrating what it takes to be academically successful; but allowing me to take ownership of my own mistakes. I thank you for your honesty and showing me that great scholarship is one who can be very critical of their own work and be open and authentic about it. To the participants of the study, I sincerely would like to thank you for your willingness to participate in such a study. Your commitment has been essential to the success of the research, and for that I am appreciative for your time and effort. All in all, it is my hope that the findings of this research will warrant more knowledge and advance this knowledge field.

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List of Abbreviations

Computers Are Social Actors (CASA)

Intelligent Personal Assistant (IPA)

Intelligent Voice Assistant (IVA)

Voice-controlled Smart Assistant (VCSA)

Chapter 1: Introduction

Overview

The problem was that there is insufficient information regarding relational and cognitive factors such as trust and perceived intelligence towards artificial intelligence (AI) voice-based assistants. The purpose of this quantitative research was to investigate how demographic variables such as generation, ethnicity, gender and education level influence perceptions of trust and perceive intelligence towards AI-voice based assistants. This chapter includes all but not limited to the quantitative research, the Background, the Problem Statement, Purpose Statement, Significance of Study, Research Questions and Hypothesis, Definition of Terms, and Summary.

Background

Technology is used to establish a real-life social presence. Traditionally, technology was used as a one-dimensional communication system to create information for the user (Standage, 1998). Today, individuals not only interact with these technological devices as tools but as social actors that can facilitate social and relational interaction. In the context of this study, social presence exists when technology users perceive media agents as human beings within the cyberspace environment. That is to say, the human experience creates a pathway for the technology experience, granting access to AI users to experience empathy, warmth and seek help with media agents. Technology no longer relies on its functional use, but rather relational use its intention to perform tasks, which has garnered more trust from its users. As far back as 2011, voice assistants were introduced as a feature a part of the iPhone 4s. Originally upon discovering computers having the ability to perform human-oriented task, John McCarthy coined the term "artificial intelligence" in 1955 due to its given performance (Siebel, 2019, p. 86). In contrast, communication scholars have argued that social responses are not geared towards the computer but rather to the programmer (Nass & Moon, 2000). Also, it has been combatted with professional researchers that a voice does not solemnly permit a social presence from technology (Reeves & Nass, 1996).

Past Situation of the Problem

In the past, research scholars have focused on how media agents send and receive online information using various types of media software agents that are produced in computer-mediated communications to execute a complex task. For example, past research showed that social cues exist in the interaction with media technologies but did not address cognitive factors such as trust and perceived intelligence (McLean & Frimpong, 2019; Nass & Moon, 2000). Originally, people only engaged with voice assistants for social purposes (Hoy, 2018) In essence, at that time voice assistants operated with built in commands and responses that was reciprocated to the user or owner with automated pre-recorded scripts. Announced in the later part of the 20th century, organizations acknowledged the importance of establishing an open and closed communication system that permitted social interaction (Ellul, 1964). Communication scholar Moon (2000) commented “reciprocity and sequence” are significant vital factors that propels people the willingness to expose information about themselves to technology (p. 323). That year, exchanging intimate information was practiced with consumers self-disclosing information to a computer. With this in mind, Stanford scholars Reeves and Nass (1996) found that people feel good when they credit themselves for the success of a computer but protect their self-esteem by blaming computers for their failures. By all means, IPA users point their finger at a computer for negative outcomes; yet its users take personal credit for when the computer produces positive performances (Moon & Nass, 1998). Leading to trust, more recently trust has been found in the adoption and usage of voice assistants (Madhavan & Wiegmann, 2007; Wu & Huang, 2021). In connection with trust and the media equation theory, since a media agent is limited in showcasing empathy the human-technology relationship is not viewed as someone who is sincere. With this intention, technology and voice assistants do not nor cannot regulate real feelings towards any user.

Current Situation of the Problem

The current situation now is the limited amount of research on the impact that generation, ethnicity, gender and education level have on user trust and perceived intelligence towards voice-based assistants (Huh et al., 2022; Nzaou et al., 2021). The belief is that voice assistants are more of a liability than asset (Bistron & Piotrowski, 2021; Vimalkumar et al., 2021). Also, that information from voice assistants is not secured and the possibility that it could be tampered with by third parties is a major concern when voice assistant users reveal personal information. Because of this, older voice-assistant users are hesitant in participating in self-disclosing acts of information to an online agent such as Apple's SIRI or Google's Alexa (Li & Rau, 2019). Not only are older generations skeptical of trusting technology that broadcasts only vocal traits; but, on the contrary, written like text responses from a computer secure and protects user emotion in compared to just the presence of a voice (Moon, 2000; Nass & Moon, 2000). So, not only has this situation not been thoroughly investigated with younger generation groups, but technology has not examined various ethnic groups in the United States who perceive voice assistants as intelligent. Moreover, few research studies have examined technology's perception of intelligence and anthropomorphism on user trust and usage of voice assistants (Moussawi et al., 2023). So, today the problem still exist that voice assistants have not been widely studied in how they are perceived intellectually and trusted by various demographic variables.

Introduction to the Problem

The general problem to be addressed is the lack of promoted knowledge on trust and perceived intelligence towards voice assistants. Hoy (2018) stated that voice assistants are limited in only helping human beings perform basic AI tasks which makes it difficult to gain not only a higher level of trust, but a greater understanding of the device. In a recent study, Ki et al. (2020) found out that a voice assistant is also acknowledged as an IPA due to its automated software application that can assist people not only with information searches but with decision making using natural language in either written or spoken form. Gillath et al. (2021) supported these views by explaining that the inability to engage in willing human-AI social interaction is directly correlated to trust. Logg et al. (2019) found that the hesitancy to warrant intelligence and full genuine trust towards a voice assistant during social interaction was by a way of perceived intelligence. Therefore, there is a challenge and gap that exist in the cybernetic human-technology experience. The specific problem to be addressed is the restricted amounts of information research committed towards how voice-based assistants are intellectually perceived and trusted by various social groups.

The increasing evolution of technology portrays human-like qualities, such as technological devices playing more of a social actor's role in everyday human being lives. In human-to-human interaction, perceived intelligence promotes intimacy and trust in creating rapport to formulate an effective cohesive relationship. Furthermore, trust creates social interaction when the device can help the user achieve a goal (Lee & Choi, 2017). Fifty-five percent of Gen Z's and millennials to early- mid-Gen X are dominant active users of smartphones that communicate with voice agents to perform social and basic tasks (Chattaraman et al., 2019; Dogra & Kaushal, 2021; Choi & Drumwright, 2021). Although early research has been consistent with the framework in the media equation theory in exploring human-computer interaction, there has been serious attention towards the computers are social actors (CASA) paradigm due to the integration of artificial intelligence applications like Siri and Microsoft Cortana to perform social and task related responsibilities for the user. In addition to the research problems, past studies have also found information exposing the problem. For example, the more satisfied one is with their technology performance the more it gets used suggesting higher levels of trust (Jo & Baek, 2023; Uysal et al., 2022). To put it differently, people evaluate trust in voice assistants the same as in human-human interaction. Additionally, recent studies have commented that trust plays a major pivotal deciding factor in the intent to use AI voice assistants (Blut et al., 2021).

Although there is an inconsistency in what gender frequency uses voice assistants to create online request, women are positively influenced more by people and things that have status-seeking titles compared to males (Liew & Tan, 2018). This research offers to close the gap between trust and perceived intelligence on how voice assistants are perceived by gender. Also, past research did not assess any social demographic variables, such as various generations, gender, ethnicity and education level whereas this research will seek to distinguish itself by granting the appropriate information.

Although this researchers plan to bridge the gap in trust and perceived intelligence on voice assistants, there is a considerable deficiency in the lack of assessed Gen Z's that are active with AI on open communication information systems. The direction for this study will seek to answer that. This study will address the deficiency by recruiting Gen Z's, various generation groups, different education levels and voice assistant users that use voice assistants for information purposes.

What are Voice Assistants

The various numbers of online systems that support voice assistants is growing at a rapid pace. With less than half of the human population using voice assistants, 16% of the world's population use voice-based assistants to complete menials everyday human tasks (Dogra & Kaushal, 2021; PWC, 2018). While this is the case, Gen Z's and Gen Alpha's are adopting the new technology the fastest and records indicate this social group are consistently active users in exercising the technology for its benefits such as making life easier (Dogra & Kaushal, 2021; Issa & Isaias, 2016; Olmsted, 2019). Voice assistants are resource tools whose main objective is to interact with computers and smartphones through dialogic interaction (Hoy, 2018; Huh et al., 2022). Popular voice assistants Apple's Siri, Microsoft's Cortana, Amazon's Alexa, and Google's Assistant represent software agents that have built-in speakers that allow for greater agent-human interaction through the use of a smartphone (Chaves & Gerosa, 2021; Poushneh, 2021). The agent actively listens to the human to process every day natural language acting as trigger points for usage activation (Hoy, 2018; Poushneh, 2021). As soon as a word cue gets appointed, the voice assistant tracks the user's voice and sends it to a network, in which then exercises and translates it as an order (Araujo, 2018; Whang & Im, 2020; Hoy, 2018). With attention to the instruction, the network grants the voice assistant with relevant knowledge to be exposed to the user, play the communication media desired by the user, or perform tasks with different online systems and tools (Hoy, 2018; PWC, 2018).

Since 2011, Apple's voice assistant Siri has been a part of the iOS community (Dasom & Kiechan, 2023; Hoy, 2018; Pal et al., 2022). Similarly, Microsoft created its own version with Cortana in 2013 (Pal et al., 2022; Hoy, 2018). In the same way, Amazon constructed Alexa in 2014, and Google's Assistant was acknowledged in 2016 (Hoy, 2018; Pal et al., 2022). Equally important, the different IPAs have their own special characteristic, but the foundations of its utility are equivalent (McLean et al., 2021; Hoy, 2018). Today, voice assistants can respond to numerous orders and questions compared to earlier voice activated technologies (Choi & Drumwright, 2021; McLean & Frimpong, 2019). As a matter of fact, human everyday language evaded irritation from the user of early voice activated technologies, which implemented exact statements where users had to match the program written language for the voice assistant to effectively communicate (Huh et al., 2022; Hoy, 2018). Therefore, to active a voice assistant one must be paired with the internet (PWS, 2018; Hoy, 2018). For that reason, each transactional message gets sent back to a networking system that evaluates the user's voice commands and supplies the voice assistant with the appropriate response (Hoy, 2018; Hsu & Lee, 2023). Historically, voice-controlled devices were only subjected to in-house commands and responses on a smaller scale that the company built in (Hasan et al., 2021; Hoy, 2018). Until now, technology has really advance in the natural language processing, recognized as "computational linguistics" (Hoy, 2018 p. 82). With this intention, it has permitted voice assistants to construct robust responses in a quickly timely manner for the user to enjoy (Hoy, 2018; Lyu et al., 2023). In like manner, research scholars acknowledge this ingenuity in natural language processing to a multitude of things (Madhavan & Wiegmann, 2007; Gambino et al., 2020). The wide range of computing power, the accessibility of large information, the intelligence of the machine and an understanding of the human language in various contexts has given the interaction between agent and human a great relationship (Gambino et al., 2020; Araujo et al., 2020; Hoy, 2018).

As technology has become more and more prevalent in everyday communication, and people have saturated the online world with heavy loads of online content to be judged, social science researchers have acquired that information to instruct voice assistants to actively listen and appropriately respond to human requests in a genuine and purposeful approach (Moussawi, 2021). Voice assistants can create requests in a multitude of different styles and make sense of what the user is seeking to explore (Zhang et al., 2023; Ahmad et al., 2022). For instance, asking Google's Assistant to recall a parked car, the user can express various phrases within the context of the question (Choi & Drumwright, 2021; Hoy, 2018). The voice assistant will locate the parked car and respond appropriately with the exact location (Moussawi et al., 2021; Hoy, 2018). The user can also generate similar related questions in the same approach requesting from the voice assistant "where is my car?" or "can you help me find my car?" (Hoy, 2018; Littlejohn et al., 2017). As a result, not only do keywords and short phrases elevate voice assistants search capabilities, but all information prompts the user to anticipate the correct response from the IPA (Ki et al., 2020; Hoy, 2018; Ortega et al., 2021).

What a Voice Assistant Does

Recently, voice assistants have showcased a new element such as "skills" whose main objective is to widen the voice assistant search capabilities by collaborating with other media agents through a web interface (Hoy, 2018 p. 83; Natale & Cooke, 2021). Nwana (1996) called this a "collaborative agent" (p. 14). Well-known voice assistant Apple's Siri has skills for connecting one to Netflix or ordering food from Uber Eats using connected information from the different networks (Olmsted, 2019). With this in mind, Zoomers and older adults have adopted in using a voice assistant due to the overall satisfaction and quality of its service (Dahlke & Ory, 2017, PWC, 2018). Subsequently, although Google's Assistant has equivalent skills it is behind in user downloads in compared to other voice assistants; so, because it was released later on as it was not equipped with the same skills as Apple's Siri (PWC, 2018; Hoy, 2018). In addition, much like how mobile apps are created for the iPhone and Android these new skills are created by independent programmers (Chaves & Gerosa, 2021 Moussawi et al., 2021). Google assistant shares data with other databases and platforms that prompts users to create their own skills (Hoy, 2018). For example, in using the voice assistant a user can set a timer to initiate a task such as create an online post like creating a tweet or even text someone at a specific time (PWC, 2018; Hoy, 2018). Researchers acknowledge this as a "web service like tasker," where one can communicate to an agent in a network that can perform and trigger other human online activities (Hoy, 2018, p. 83; Chen & Park, 2021). To elaborate, greeting an IPA like Apple's Siri can initiate other behaviors such as turning on the lights or locking the doors to increase a users' consistency of daily behavior (PWC, 2018; Hoy, 2018). The connection with the devices is that using a voice assistant closes the gap that encourages users to express commands orally than through a click on an app.

How Voice Assistants Are Being Used in the Home

Voice assistants are accessible on a plethora of connected devices on a given network (Hoy, 2018; PWC, 2018). Apple, Google and Amazon employ home speakers for the voice assistant to perform daily functions mostly controlling smart online connected devices (Wienrich et al., 2023; Ortega et al., 2021). In addition, Hoy (2018) addressed that Amazon's Echo has numerous versions that can demonstrate not only audio commands but produce video content as well. As explained by researchers, Google's speaker is referred to as "Home" because it's main activity controls in-house operations like the thermostat (Hoy, 2018, p. 82; McLean & Frimpong, 2019). Not to mention, the speaker comes in various sizes depending on the preference of the user (Dasom & Kiechan, 2023; Hoy, 2018). Since Apple's announcement of the HomePod in 2017 it has prompted their counterparts like Microsoft